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List of StakeHolders DOCX

List of Stakeholders

A list of stakeholders or entities that have an interest in the outcome of the project or problem. Columns (interest), (Power) have 2 options High,Low. Power indicates how much a stakeholder can influence the outcome of the project. Interest is how closely the stakeholder involved with the project.

SK_ List of Stakeholders

Stakeholder ID	Stakeholder Name	Role/Position	Category	Contact Information	What's important to the Stakeholder (Interest)	Level of Power/Influence	Level of Interest
SK.1	Toyota	Innovation Dept.	Sponsor	NA	The project is an initiative of the Innovation department at Toyota, hnece they are direct owners/sponsors of the project.	High	High
SK.2	NA	Inspection Manager	Beneficiaries	NA	Supervise the inspection officer's work.	High	High
SK.3	NA	Inspection Officer	User	NA	Use the glasses for inspecting vehicles on the plant.	Low	High
SK.4	NA	Data Analyst	Integrators	NA	Use the Data received form the glasses to analyze and generate reports accordingly.	Low	High
SK.5	NA	Safety Officer	Safety & Compliance	NA	Ensure the smart glasses comply with workplace safety standards and do not introduce new hazards (distraction, electrical risk, visibility obstruction, ergonomic strain).	High	Low
SK.6	NA	Compliance Auditor	Safety & Compliance	NA	Ensure inspection processes and recorded data comply with regulatory standards, company policies, and audit traceability requirements.	High	Low
SK.7	NA	Assembly Line Worker	Affected Non-User	NA	Maintain smooth assembly line operations.	Low	Low
SK.8	NA	IT Support Staff	Maintainers	NA	Ensure system reliability, network connectivity, data security, and ease of maintenance.	Low	Low
SK.9	NA	Equipment Maintenance Technician	Maintainers	NA	Maintenance of the glasses, after and before use. Direct interest.	Low	Low

List of Stakeholders 1

List of Stakeholders 2

Raw Evidence Docs:

Name	Date modified	Type	Size
Audio Recording 4-2-2026	16/02/2026 11:48 pm	Microsoft Word D...	13 KB
Huddle 3-2-2026	16/02/2026 11:55 pm	Microsoft Word D...	13 KB
Huddle 30-1-2026	16/02/2026 11:54 pm	Microsoft Word D...	13 KB
Sheeba and Zainab (recording) transcript...	16/02/2026 11:53 pm	Microsoft Word D...	14 KB
Transcription of 1st Meeting with Toyota ...	16/02/2026 11:53 pm	Microsoft Word D...	15 KB
VideoAudio Recording 4-2-26	16/02/2026 11:46 pm	Microsoft Word D...	13 KB

The Docs do not contain any written data here (except the descriptions), since the actual recordings are uploaded with the artifact on Innoslate. Which is shown in screenshots further.

The transcriptions are in statement format hence it is available in the description:

MM.4.(Team + PM + Toyota Team)

MM.4.(Team + PM + Toyota Team) Transcription of 1st Meeting with Toyota (Zainab + Areeb)

Project Objectives & Scope
• Primary Goal: The main objective is to make the existing eye-tracking hardware functional for inspection purposes. The system needs to map the operator's retina movement and generate a heat map to verify where they are looking.
• Core Function: The system must ensure "hand-eye coordination" during assembly tasks (e.g., tightening bolts, installing relays). It verifies that the operator is actually looking at the specific part they are working on, reducing human error in manual assembly.
• Deliverable: A workable prototype that links software to the existing hardware. The stakeholders emphasized that they do not expect an industrial-grade product immediately, but rather a functional baseline.
Technical Approach
• Hardware vs. Software:
• Hardware: The team is advised not to build new glasses/hardware from scratch, as that is a massive undertaking (estimated at 10 years/post-doc level work). Instead, they should utilize the existing eye-tracking device provided by Toyota.
• Software: The students' primary focus should be on the software side—building the system architecture, processing data, and creating the interface to map the retina tracking and heat maps.
• Connectivity: A key task is to establish connectivity with the existing device (checking if it uses Bluetooth, Wi-Fi, etc.) to extract data for the new software system.
• Ambition Level: The stakeholder (Areeb) explicitly warned against being too ambitious (e.g., adding complex AI/Machine Learning immediately). The priority is to first achieve the basic functionality of tracking and mapping. AI can be a future milestone.
Use Cases in Toyota (IMC)
• Inspection: To check compliance with Standardized Work (SW) or Work Instruction Sheets (WIS), especially in hidden or difficult-to-see areas like the engine room or vehicle interiors.
• Process Audit: Reducing the time required for audits. Currently, manual checks take a long time; the device could reduce a 1-hour task to 10-15 minutes.
• Safety & Quality: Ensuring operators are focused on the correct components to prevent defects.
Timeline & Management
• Immediate Next Steps: The students will inspect the provided hardware to understand its connectivity and limitations.
• Meeting Cadence: Progress meetings are suggested to occur approximately twice a month.
• Team Composition:
• Sheeba: Project Manager.
• Bisma: Project Coordinator from Habib University.
• Student Team: Divided into roles including Computer Vision/Estimation (Shamsi), Analytics/Data Processing (Vasik), and others handling research and embedded systems.
• Toyota Stakeholders: Zainab and Areeb (Arif).

MM.3.(PM + Zainab)

MM.3.(PM + Zainab) Sheeba and Zainab (recording) transcription

Based on the meeting transcript provided, here are the notes regarding the project clarification between Sheeba and Zainab:
• Clarification from Stakeholder: The stakeholder (Areeb) clarified that the motive is to utilize frames rather than just the glasses.
• Reference Material: Research papers and data from a previous startup that worked on eye tracking glasses will be shared with the students. This will help them understand the domain and necessary study materials.
• Known Issues: It was noted that the previous attempt faced persistent connectivity or system issues despite continuous work, which the students should be aware of.
• Immediate Tasks: The students are expected to spend the next few months on requirement gathering, analysis, and system architecture to determine the best design and system.
• Key Deadline: They must submit a proposal and give a presentation by the end of April.
• Development Phase: Actual work on the system and software development will begin after the April presentation. It was confirmed that software development would start first while the hardware remains the same.
• Hardware: The project (referred to as SVR) does not require extensive hardware. It may require cameras, which are typically provided for students, so no significant hardware expenses are anticipated.
• Progress Tracking: Online meetings will be set up with the students every two weeks to track their progress.
• Zainab's Role: Zainab is responsible for tracking progress in these meetings, keeping the students motivated, and ensuring they deliver on what they promise for each two-week sprint.
• Documentation: Bisma will share the meeting minutes.

User Needs Document:

User Needs Document

UND.1 Project Overview

Commercial eye-tracking systems used in manufacturing are expensive, import-dependent, and internet-reliant, making them difficult to deploy on plant sites due to cost, connectivity, and data-security constraints. Many low-cost alternatives do not detect actual pupil movement, leading to inaccurate conclusions about where an operator is truly looking. There is a need for a cost-effective, locally developed, offline eye-tracking solution that can be safely used within manufacturing plants and allows post-task diagnosis through recorded video analysis.

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.1 Capture the vehicle completely

The Inspection Officer needs to capture exactly where they are looking onthe vehicle body and interior during an inspection.

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.2 1st person video recording

The Inspection Officer needs to record a first-person video view of thevehicle being inspected to provide context for the gaze data.

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.3 Eye Sequence Captured

The Inspection Manager needs to visualize the sequence and flow of theofficer's inspection (e.g., where they started, where they moved next).

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.4 Every action captured

The Inspection Manager needs to determine if the Inspection Officerskipped any required steps or areas during the standard inspectionprocess.

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.5 Visual focus mapping

The Data Analyst needs to identify which areas of the car received the mostvisual attention and which were neglected.

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.6 Audit time reduction

The Industry needs to reduce the time required for process auditssignificantly (targeting a reduction from 45 mins to 10-15 mins).

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.7 100% defect-free cars

The Inspection Manager needs to ensure that zero defective cars leave theinspection area by guaranteeing 100% of the inspection zones are visuallychecked.

Rationale:

Quality Score: 0%

Labels: *None to display.*

UND.1.8 Accurate sequence of actions captured

The Inspection Manager needs to display the exact sequence of a correctinspection to trainees using visual maps and videos, rather than verbal andphysical demonstrations, to improve training efficiency.

Rationale:

Quality Score: 0%

Labels: *None to display.*

End of document

Artifacts Created:

SR.0 Stakeholder Requirements	Requirements Document	5605179
AUD.2.(PM + Team + Zainab + Areeb Bhai) Video/Audio Recording 4-2-26	*No labels to display.*	5615655
AUD.1.(PM +Zainab) Audio Recording 4-2-2026	*No labels to display.*	5615029
UND.0 User Needs Document	Requirements Document	5615680
SD.0 List of Stakeholders	Notes Document	5617855
VR.1 Stakeholder Verification Requirements	Requirements Document	5617958
MM.2.PM Huddle 3-2-2026	*No labels to display.*	5605924
MM.1.(PM + Assistant PM) Huddle 30-1-2026	*No labels to display.*	5605526